A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 17

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) **1** These react with oxygen in the air spaces between the waste particles (oxidation) **1** Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons **1** Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) **1** Only a few tolerant taxa survive – the sensitive mayfly larvae (indicator of clean water) are eliminated – dissolved oxygen is not the limiting factor here **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dilution by tributaries / more water **1** Neutralisation by alkaline rocks (limestone) in the stream bed – zinc precipitates / adsorbs onto sediments as pH rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid **1** Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** A turbofan at the front forces bypass air around the inner cowling / core jet **1** The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise **1** The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly **1** Acoustic liners in the engine cowlings absorb noise **1** Hush kits fitted to older low-bypass engines **1** Fairings on undercarriage and wing-flap hinges reduce turbulence – welded rather than riveted panels **1** Lighter composite aircraft need less thrust **1** Blended wing with engines on top so the body shields the ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer **1** Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low **1** Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft **1** More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
02.4	1 mark for the point below. **1** Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	

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Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Mann-Whitney U test **1** Chi-squared test **1** t-test **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chi-squared compares frequencies / counts in categories **1** Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **chi-squared** → a test comparing observed frequencies with expected frequencies • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is always some remaining uncertainty / a probability that the result was produced by chance **1** Statistics only give a level of confidence (eg 95%, 99%) that the result is significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bars drawn above and below each mean showing plus and minus one standard deviation **1** If the bars of two means overlap a lot the difference is probably not significant – if they do not overlap it probably is **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 **1** Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** 24-hour day / night cycle: people asleep overnight – morning rise as heating, lighting, cooking, industry and offices start **up to 2 Evening peak:** lighting, cooking, heating as people return home – short-term 'TV pickup' at programme breaks / mealtimes **up to 2 Other causes:** seasonal (winter heating / lighting, summer air conditioning) – weather-related (cold snaps) – weekday / weekend differences in industrial use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly **1** Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Intermittent renewable resources:** wind, solar, tidal **1** Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) **1** Breakdowns / maintenance of power stations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic **1** Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it – the cold water returns southwards at depth (thermohaline circulation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** Higher temperatures melt land ice on Greenland which flows into the North Atlantic **1** The fresh water dilutes the seawater, reducing its salinity and density **1** Less dense water sinks less easily so the flow rate of the current is reduced **1** Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them **1** Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change – individual events do not show trends **1** Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity) – incomplete understanding of the processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
05.4	1 mark for the point below. **1** Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data – the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1 Fission:** splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons **1 Fusion:** joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Deuterium (hydrogen-2) extracted from (sea)water **1** Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide **1** Vacuum so the plasma is not cooled by air **1** Magnetic field holds the plasma centrally so it does not touch the container walls and cool **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Fuel (deuterium / lithium) is abundant / effectively unlimited **1** Much less radioactive waste (no long-lived fission products) – no chain reaction / meltdown risk – no weapons material **1** Not yet achieved sustained net energy output: reactors (JET) use more energy than they release – ITER aims for 500 MW out from 50 MW in – laser fusion also experimental – commercial viability decades away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	
08.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) **1** Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles **1 Carbon neutral in principle:** CO2 released equals CO2 absorbed in photosynthesis **1** Energy density of alcohol / biodiesel nearly as high as fossil fuels – wastes (straw, sewage, landfill gas) are cheap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **landfill gas** → the methane and carbon dioxide produced by anaerobic decomposition of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) **1** Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO2 as using fossil fuels directly **1** Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally **1** Supply from wastes limited to the amount produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Hydrogen from algae:** some algae deprived of sulfur produce hydrogen during photosynthesis **1** Anaerobic digestion extended to crop wastes and marine algae – small-scale digesters (eg one cow's manure supplying a family's cooking gas) **1** Biofuels from microorganisms (eg oil-producing microalgae) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK **1** New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) **1** Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) **1** Lighter designs using less material – longer vehicle life spreading the embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Use of recyclable materials wherever possible **1** Easy identification of components and their composition (code stamps) **1** Easy dismantling and separation of components / materials **1** Reusable components for new vehicles **1** Compostable materials for parts that cannot be recycled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** At low speed a large proportion of the energy just runs the engine rather than moving the car **1** At high speed energy is wasted overcoming air resistance (which rises with the square of speed) – optimum for many cars about 56 mph **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	1 mark for the point below. **1** Automatic stop / start systems cutting the engine when stationary – smooth driving in higher gears avoiding sudden braking / acceleration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** Lasky's principle - vast quantities exist at low grade – reserves vs resource – COOG affected by price, technology, energy costs • **Improved exploration:** higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • **Low-energy extraction of low-grade ores:** bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • **Previously inaccessible deposits:** polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • **Extending use:** recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • **Evaluation:** environmental impacts vs benefits – energy costs – economics – which strategies are most sustainable – conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **recycling** → reprocessing a material to make something new	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Lasky's principle means that as grade falls, the quantity of metal available rises enormously, so depletion of high-grade deposits does not mean running out - it means the cut-off ore grade must fall, which depends on price, technology and energy costs. Better exploration finds what remains: high-resolution multispectral and infrared satellite sensors, gravimetry, magnetometry, seismic and resistivity surveys and portable field analysers narrow the search so expensive trial drilling is targeted. Mechanisation allows deeper underground mines and larger open-cast pits, and previously inaccessible deposits - polymetallic nodules and deep-sea sediments, and Antarctica if its protection were ever lifted - become possible, with serious environmental and legal problems.	
	The most sustainable route to low-grade ores is low-energy extraction: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement of copper, bacterial adsorption of rare earths and polymer adsorption from seawater all avoid the energy of crushing and smelting vast tonnages. Equally important is extending the use of metal already mined: recycling, Cradle-to-Cradle design so products can be disassembled, longer product lifetimes and substitution of scarce metals - the circular economy approach.	
	Each strategy has costs: deep-sea mining and Antarctica risk unique habitats, deeper mines need more energy, recycling is limited by dispersal and mixed alloys. The strategies that mimic natural low-temperature processes and keep metals in use are the most sustainable, and exploration and new mining should be the last resort rather than the first.	
	Weaker answer - Level 1 (3 marks)	
	When high grade ores run out we can mine lower grade ores instead, but this uses more energy. Better technology like satellites can find new deposits. We could mine the deep sea or Antarctica. Bacteria can be used to extract metals from low grade ore. Recycling metals means we do not need to mine as much. These strategies will help secure supplies of metals in the future.	
	What makes the difference	
	• Level 1 names most strategy types but Level 3 explains each (what the surveys measure, why low-temperature extraction matters, what Lasky's principle implies).	
	• Level 3 weighs environmental and energy costs and ranks the strategies; Level 1 offers no evaluation.	
	• Framing the answer with the concept of cut-off ore grade shows understanding rather than recall.	
10.2	1 mark for the point below. **1** All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

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